MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 5667

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, forced-air cooled

See figure 1

Weight: 40-pounds (18.14 kg) nominal

Mounting position: Vertical, anode down

ABSOLUTE-RATINGS: C Telegraphy

Parameter:	Ef	Eb	Ib	Pi	Pp	Ec	Ic	Radiator T	Glass T
Unit:	V	kVdc	Adc	kW	kW	kVdc	Adc	°C (see note 1)	°C
Maximum:	11.55	10.0	2.0	20.0	7.5	-1.5	0.35	230	160
Minimum:	10.45	---	---	---	---	---	---	---	---
TEST CONDITIONS:	11.0 Vac	7.5	---	---	---	---	---	See note 2	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	119	128	Adc
1266	Total grid current	5	Eb = 10 kVdc; Ec/Ib = 1.0 Adc, t = 300 sec	0.65	II	Ic	---	-100	μAdc
1261	Electrode voltage (1) (anode)	-	Eb/Ib = 1.0 Adc; Ec = -200 Vdc	0.65	II	Eb(1)	6.50	8.50	kVdc
1261	Electrode voltage (2) (anode)	-	Eb/Ib = 1.0 Adc; Ec = 0	0.65	II	Eb(2)	2.80	3.80	kVdc
1261	Electrode voltage (grid)	-	Ec/Ib = 20 mAdc	0.65	II	Ec	---	-475	Vdc
1231	Emission	-	Ef = 8 Vac; Eb = Ec = 1.0 kVdc	0.65	II	Ic	0.50	1.40	Adc
1236	Power oscillation	-	Eb = 14 kVdc; Ib = 2.0 Adc; Rg = 6,000 ohms; Ic = 0.300 Adc; F = 1.5 MHz; t = 600 sec	0.65	II	Po	18	---	kW (useful)
1266	Primary grid emission	-	Ic = 1.15 Adc; Eb = -2.0 kVdc	0.65	II	Isg	---	-500	μAdc
	Quality conformance inspection, part 2								
---	Terminal deflection	4		---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-	No shield	---	---	Cgp Cin Cout	15.8 19.2 2.0	21.2 27.4 4.0	pF pF pF
1316	Amplification factor	-	Eb(1)-Eb(2)/0.20	---	---	Mu	18	24	---
	Quality conformance inspection, part 3								
---	Service-life guarantee	-	t = 2,000 hours	---	---	---	---	---	---
---	Terminal torsion	3,6		---	---	---	---	---	---

NOTES:

1. Airflow of at least 550 cfm, at a minimum pressure of 0.8 inches of water shall be flowing before application of any voltages. Airflow of approximately 15 cfm through a 3-inch (76.2 mm) diameter nozzle directed at the upper part of the bulb is required to limit temperature of glass at hottest point of 160°C.

2. The airflow shall not exceed 550 cfm at 0.8 inch of water. The upper bulb cooling should not exceed 15 cfm from a 3 inch (76.2 mm) nozzle. At the option of the manufacturer, type 5667 tubes may be tested according to this specification as a water-cooled tube prior to assembly with the radiator, in which case only test method 1266 (total grid current) need be made on the assembled 5667 tube. A holding period of 24 hours minimum (after attaching radiator) shall be made prior to making test method 1266.

3. Terminal torsion. The tube will be secured horizontally with the two filament terminals in a vertical plane and an aluminum bar, Drawing 156-JAN, clamped in a horizontal position to top filament terminal. There will be suspended without shock at 10 inches (254.0 mm) from the center of the terminal a weight of 10 pounds (100 inch-pounds, neglecting weight of aluminum bar) for a period of at least 5 minutes.

   (a) There should be no evidence of failure in seal nor shorting of electrodes. This test shall be conducted prior to electrical tests.

   (b) The tube will be secured horizontally with the two grid terminals in a vertical plane and the foregoing test repeated on the top grid terminal.

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NOTES: -Continued

4. Terminal deflection. The tube will be secured in the normal operating position and an aluminum bar, Drawing 156-JAN, clamped parallel with the plane of the grid terminals to one filament terminal. A weight of 12.5 pounds (125 inch-pounds, neglecting weight of aluminum bar) will be suspended without shock at 10 inches (254.0 mm) from the center of the terminal for a period of at least 5 minutes. There should be no evidence of failure in seal, distortion of collar, nor shorting of electrodes. The foregoing test will be repeated on one grid terminal, the clamped bar being parallel with the plane of the filament terminals.

5. This test is to be the first test performed at the conclusion of the holding period.

(C) 6. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

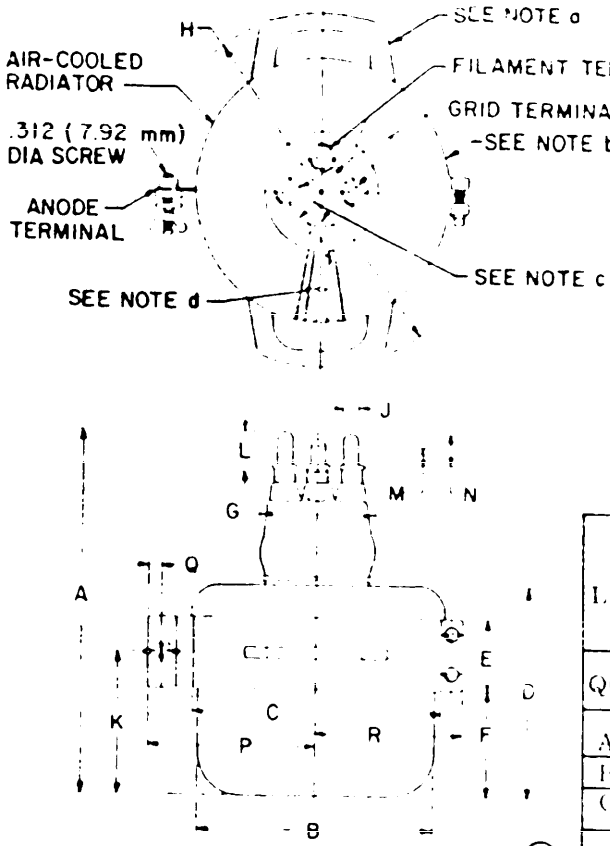
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2928)

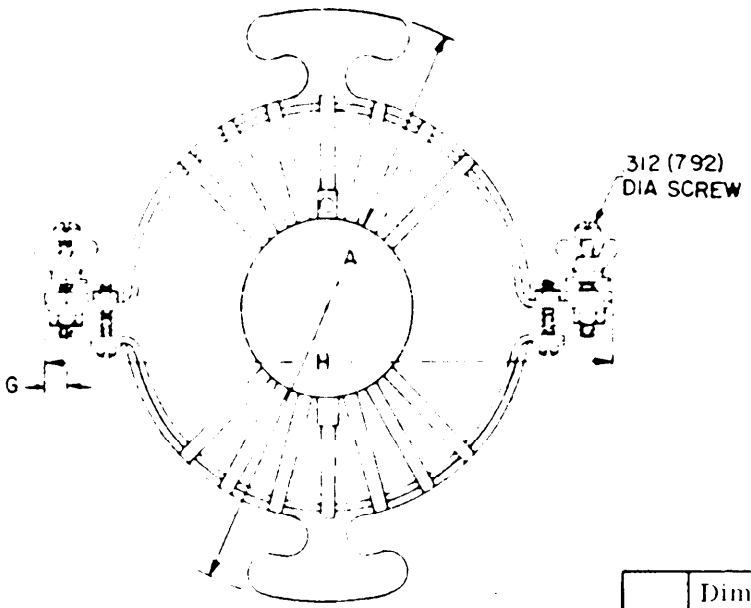
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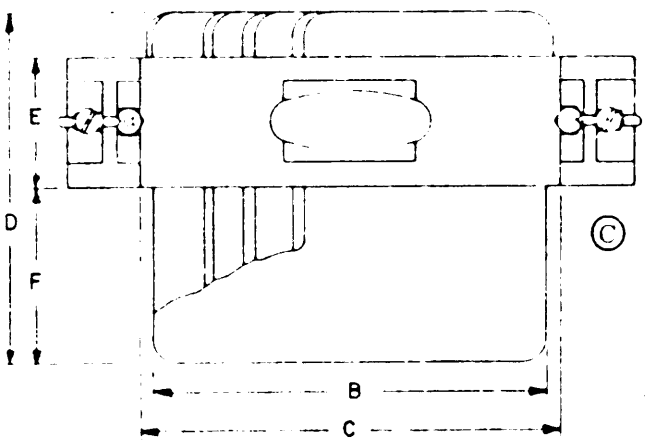


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	11.750 (298.45)
B	7.438 (188.93)	7.563 (192.10)
G	---	3.625 (92.08)
Ⓒ Quality conformance inspection, part 2 (see note e)		
C	7.875 (200.03)	8.125 (206.38)
D	---	6.563 (166.70)
E	2.188 (55.58)	2.563 (65.10)
F	3.125 (79.38)	3.375 (85.73)
H	10.500 (266.70)	11.000 (279.40)
J	.432 (10.97)	.442 (11.23)
K	---	4.375 (111.13)
L	.750 (19.05)	---
M	.465 (11.81)	---
N	.650 (16.51)	---
P	5.250 (133.35)	5.500 (139.70)
Q	.406 (10.31)	.469 (11.91)
R	---	5.500 (139.70)

FIGURE 1. Outline drawing of electron tube type 5667.

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ALTERNATE OUTLINE DRAWING OF RADIATOR

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
B	7.438 (188.93)	7.563 (192.10)
G	.375 (9.53)	.500 (12.70)
Quality conformance inspection, part 3 (see note e)		
A	10.563 (268.30)	10.938 (277.83)
C	7.875 (200.03)	8.125 (206.38)
D	6.375 (161.93)	6.625 (168.28)
E	2.188 (55.58)	2.563 (65.10)
F	3.125 (79.38)	3.375 (85.73)
H	10.563 (268.30)	10.938 (277.83)

NOTES:

a. Radiator handles must be in line with filament terminal.
b. As alternate, the radiator shown may be used.
c. The tube base must enter to a distance of .625 inch (15.88 mm) into a flat gage having 4 holes .536 ±.001 dia (13.59 - 13.64 mm) on a 2.125 ±.001 dia (53.95 - 54.00 mm) B C at angles of 90° ±10°.
d. Thermometer well optional, must be in line with filament terminals.
e. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 5667 - Continued.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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